TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

OCTOBER EXAMINATION

# BMSE3153 SOFTWARE PROJECT MANAGEMENT

FRIDAY, 3 OCTOBER 2025 TIME: 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN INFORMATION SECURITY
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS DEVELOPMENT

BACHELOR OF SOFTWARE ENGINEERING (HONOURS)

**Instructions to Candidates:**

Answer **ALL** questions. All questions carry equal marks.

The Customer Service Department of ABC Bank aims to enhance queue and appointment management at service counters by shifting from a paper-based ticketing system to an online queuing solution. You have been assigned to develop an e-Queuing System (eQS) for all its branches. The user gets a queue number by scanning a QR code at the branch office. The eQS will inform the users of their queue number and expected waiting period to be served at the service counter. The users receive notifications about their upcoming turns and appointment reminders.

A budget of RM 155,000 has been approved and this project must be completed within 3 months. To achieve a successful outcome, the project team must consistently follow software quality management, risk management, and software process improvement practices throughout the project's duration. The scope of the project focuses on the essential features of the three modules: queuing, time tracking, and notifications.

## Question 1

Answer the following questions based on the given scenario.

a) A network diagram can be used to illustrate the interdependencies between tasks in a project. Table 1 shows the tasks, durations and predecessors for the eQS project.

Table 1: Project tasks, duration and its predecessor

| Task ID | Task Name | Duration (in weeks) | Predecessor |
|---|---|---|---|
| A | Proposal | 1 | None |
| B | Requirements analysis | 1 | A |
| C | Design UI and data models | 2 | B |
| D | Design an initial prototype | 2 | B, C |
| E | Code on modules | 1 | D |
| F | System testing | 2 | E |
| G | Documentation | 1 | B |

(i) Create a network diagram based on the task ID, duration and predecessors provided in Table 1. (8 marks)

(ii) Based on the answer given in Question 1 a) (i), identify the critical path by analysing task IDs and their dependencies, specify the total duration needed to complete the entire project. (4 marks)

b) Maintainability and correctness are two quality attributes in McCall's software quality model. Select a suitable quality attribute for eQS's project development. Explain the selected attribute and analyse **ONE (1)** metric that can be used to measure it. (7 marks)

c) The eQS project's development process incorporates both Quality Assurance (QA) and Quality Control (QC) practices. Explain how each contributes in improving the software quality. (6 marks)

[Total: 25 marks]

## Question 2

Answer the following questions based on the given scenario.

a) Table 2 shows the current progress of project eQS with a total allocated budget of RM 155,000. As the project is now in its seventh week, the planned value reflects the estimated cost of tasks scheduled for completion by week 7 and the actual cost (AC) incurred is RM 114,000.

Table 2: Project task progress, duration, estimated cost and completion status.

| Task ID | Estimated Duration (weeks) | Estimated Cost (RM) | Completion Status (%) |
|---|---|---|---|
| A | 1 | 10,000 | 100% |
| B | 1 | 20,000 | 100% |
| C | 2 | 30,000 | 100% |
| D | 2 | 20,000 | 100% |
| E | 1 | 40,000 | 90% |
| F | 2 | 20,000 | 0% |
| G | 1 | 15,000 | 0% |
| Total | 10 | 155,00 | - |

Note:
- Show the working in your calculation. Formula used: SV=EV-PV, CV=EV-AC.

(i) Calculate the following values based on the information provided in Table 2:
- Planned Value (PV) (2 marks)
- Earned Value (EV) (4 marks)
- Schedule Variance (SV) (2 marks)
- Cost Variance (CV) (2 marks)

(ii) Based on the results of your calculations in Question 2 a) (i):
- Provide comments on the performance of this project.
- Analyse **ONE (1)** corrective action to ensure this project's success.

(4 + 4 marks)

b) Challenges in implementing process change within software process improvement initiatives include resistance to change and maintaining change persistence. Choose **ONE (1)** potential challenge that may arise during eQS's project implementation. Explain **TWO (2)** reasons to support your choice. (7 marks)

[Total: 25 marks]

**Question 3**

Answer the following questions based on the given scenario.

a) Project manager has identified some potential risks of the eQS project in Table 3.

Table 3: Project risk, probability, impact and risk score

| No. | Risks | Probability of occurrence (0 to 1) | Impact on the project (1 to 4) | Risk score |
|---|---|---|---|---|
| 1 | Team members are resigned | 0.2 | 2 | |
| 2 | Constant changes in requirements | 0.4 | 2 | |
| 3 | Junior developers unable to complete modules on time | 0.6 | 3 | |
| 4 | Lack of teamwork | 0.5 | 2 | |

Note:
- Risk score metric is probability X impact
- Impact values: 1 is negligible, 2 is marginal, 3 is critical and 4 is catastrophic.

(i) Calculate the *risk score* for each risk listed in the last column of Table 3. Rank the risks according to their priority using risk No. in the first column. (i.e. from highest to lowest risk score). (7 marks)

(ii) Based on the results of your calculation in Question 3 a) (i), identify **ONE (1)** risk with the highest risk score. Analyse a suitable risk mitigation, monitoring and management actions for the identified highest risk. Explain your answer. (10 marks)

b) The project manager has decided to use the Centralised Version Control (CVC) system in eQS's project development. Explain **TWO (2)** features of CVC system that the project team can utilise effectively and give **ONE (1)** example of CVC system. (8 marks)

[Total: 25 marks]

## Question 4

a) Table 4 provides size-oriented measurements for Project A and Project B.

Table 4: Number of lines of code, documentation pages and cost for both projects.

| | Project A | Project B |
|---|---|---|
| Lines of code (LOC) | 125,000 | 160,000 |
| Documentation pages | 165 | 152 |
| Cost (RM) | 150,000 | 192,000 |

Note:
- Assume that both projects use the same programming language and same project scopes.
- Round up your answer to three decimal points.
- KLOC is 1,000 lines of code

(i) Using the normalised size-oriented metric, calculates the following values for each project based on the information provided in Table 4:
- Documentation pages per 1,000 lines of code
- Cost per lines of code
- Cost per documentation pages

(6 marks)

(ii) Based on the results of your calculations in Question 4 a) (i), give an analysis of which project has higher *maintainability* and incurs *lower cost* in producing documentation. Explain your answer. (5 marks)

b) *Availability* and *reliability* are essential dimensions of a Cash Deposit Machine's system dependability. Explain how availability influences the machine's reliability from the customer's perspective. (6 marks)

c) *Waterfall* and *Rapid Application Development* software process models are commonly used in project development. Analyse **TWO (2)** factors to be considered when selecting between these models in software projects development. Explain your answer. (8 marks)

[Total: 25 marks]